

Customer Churn EDA Report

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Dataset: Telco Customer Churn (IBM Sample Data Sets)

1. Introduction

Customer churn, also known as customer attrition, refers to the phenomenon where customers stop doing business with a company. In the telecommunications industry, churn is a critical business metric because acquiring new customers is significantly more expensive than retaining existing ones. Studies show that increasing customer retention rates by just 5% can increase profits by 25% to 95%.

This report presents a comprehensive exploratory data analysis (EDA) of a telecom customer churn dataset. The primary objectives of this analysis are:

1. Understand the patterns and factors that influence customer churn.
2. Identify high-risk customer groups.
3. Detect data quality issues and outliers.
4. Provide actionable business recommendations to reduce churn rates.

The analysis follows a structured approach including data cleaning, univariate analysis, bivariate analysis, and correlation analysis to uncover meaningful insights from the data.

2. Dataset Details

Dataset Information

Aspect	Details
Dataset Name	Telco Customer Churn
Dataset Source	Kaggle (IBM Sample Data Sets)
Number of Rows (Original)	7,043

Aspect	Details
Number of Rows (After Cleaning)	7,021
Number of Columns (Original)	21
Number of Columns (After Cleaning)	20
Target Column	Churn
Overall Churn Rate	26.45%

Feature Types

Feature Type	Columns
Numerical Features	tenure, MonthlyCharges, TotalCharges
Categorical Features	gender, SeniorCitizen, Partner, Dependents, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, PaymentMethod

Dataset Description

Each row in this dataset represents a unique customer. The columns contain comprehensive customer attributes including demographic information, account details, and service subscriptions. The target column Churn is a binary variable indicating whether a customer left the company within the last month (Yes) or not (No).

3. Data Cleaning Summary

The following data cleaning steps were performed to ensure data quality and prepare the dataset for analysis:

Cleaning Task	Action Taken	Result
Missing Values	Checked for null values in all columns	No missing values found (0% missing)

Cleaning Task	Action Taken	Result
Duplicate Records	Identified and removed duplicate rows	22 duplicate rows removed
Wrong Data Types	Converted TotalCharges from object to float64	Correct data type for numerical analysis
Wrong Data Types	Converted SeniorCitizen from int64 to category	Proper categorical representation
Unnecessary Columns	Removed customerID column	Dropped as it's not useful for analysis
Inconsistent Values	Stripped extra spaces from string columns	Clean and consistent string values
Outliers	Detected using IQR method and boxplots	No outliers detected in any numerical column

Data Quality Assessment Results

Quality Metric	Result
Missing Values	0% (No missing values)
Duplicate Rows	22 removed
Outliers	None detected
Data Types	All corrected
Final Dataset Shape	7,021 rows × 20 columns

The dataset is of high quality with no missing values, no outliers, and all features correctly typed for analysis.

4. Exploratory Data Analysis

4.1 Univariate Analysis

Univariate analysis examines each column individually to understand its distribution and characteristics.

4.1.1 Churn Distribution

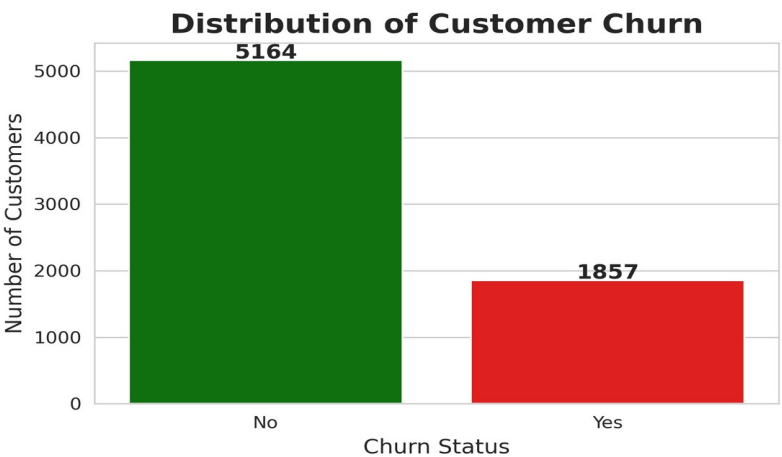


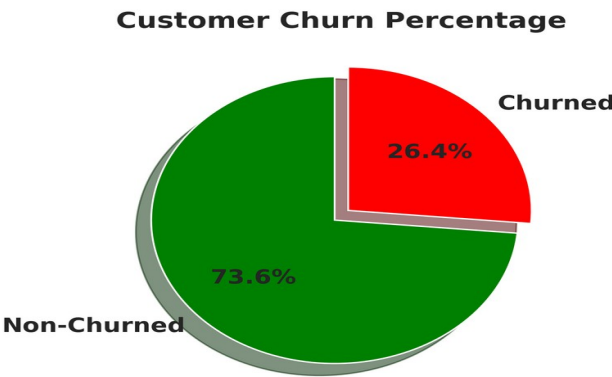
Fig1: Churn Count Plot

Churn Status	Count	Percentage
Non-Churned	5,164	73.55%
Churned	1,857	26.45%

The analysis shows that 26.45% of customers (1,857 out of 7,021) have churned, while 73.55% (5,164 customers) remain active. This indicates a significant churn problem that requires immediate attention.

Observation: The churn rate of 26.45% is relatively high for the telecommunications industry, suggesting that customer retention should be a top priority

Fig2:Churn Pie Plot



The pie chart visually represents the churn ratio, showing that more than a quarter of customers have left the company.

4.1.2 Tenure Distribution

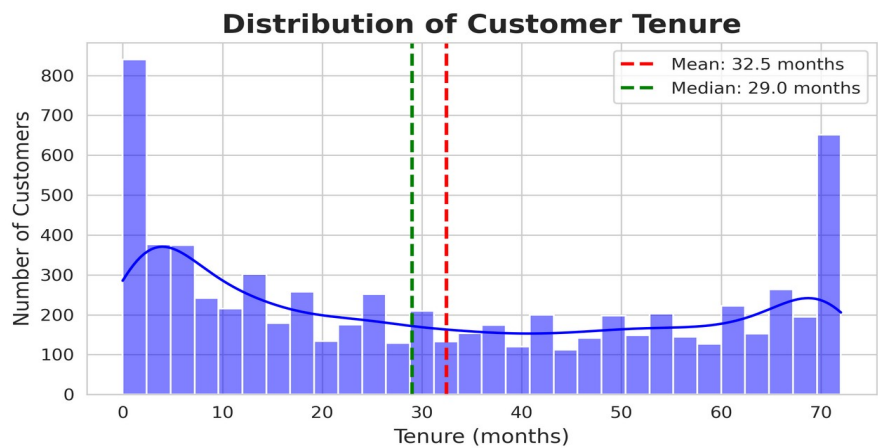


Fig3: Tenure Distribution Histogram

Statistic	Value
Mean Tenure	32.47 months (~2.7 years)
Median Tenure	29.00 months
Minimum Tenure	0 months
Maximum Tenure	72 months
Standard Deviation	24.53 months

Observation: The tenure distribution shows a peak at very low tenure (new customers), with a steady decline. This suggests that new customers are at higher risk of churning, which is a common pattern in subscription-based businesses.

4.1.3 Monthly Charges Distribution

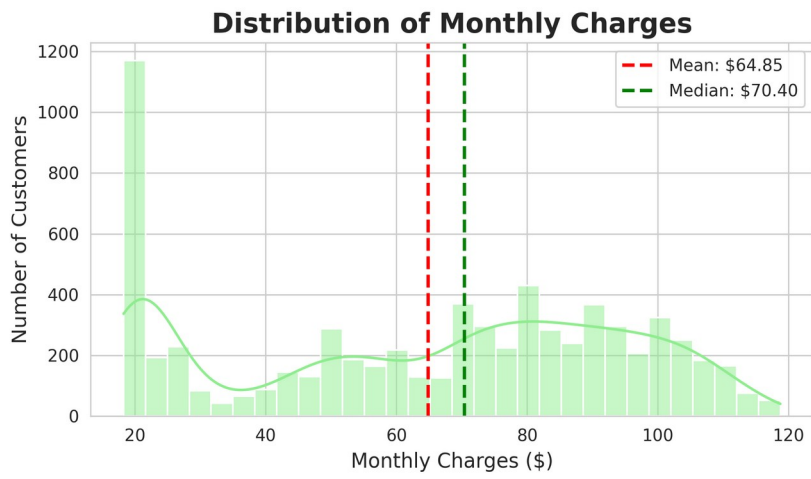


Fig4: Monthly Charges Histogram

Statistic	Value
Mean Monthly Charges	\$64.85
Median Monthly Charges	\$70.40
Minimum Monthly Charges	\$18.25
Maximum Monthly Charges	\$118.75
Standard Deviation	\$30.07

Observation: The distribution of monthly charges is relatively spread out, with a slight peak around \$70-90. There is a wide range of pricing, indicating diverse customer segments.

4.1.4 Senior Citizen Distribution

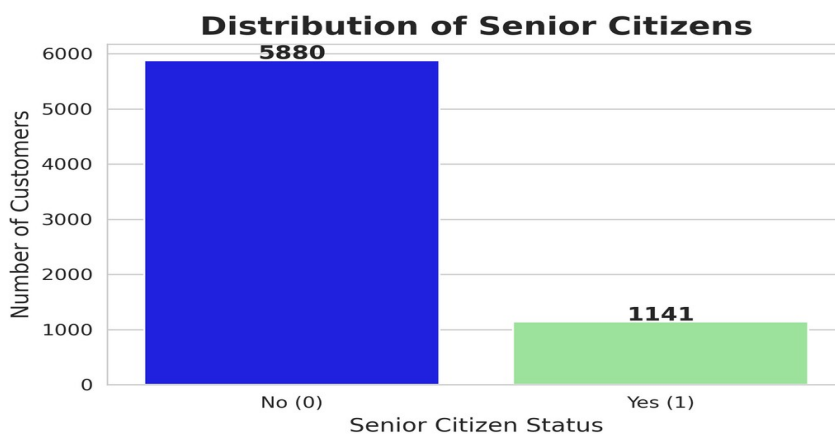


Fig5:Senior Citizen Distribution Countplo

Category	Count	Percentage
Non-Senior Citizen	5,880	83.75%
Senior Citizen	1,141	16.25%

Observation: Senior citizens represent a minority of the customer base but show a significantly higher churn rate as seen in the bivariate analysis.

4.2 Bivariate Analysis

Bivariate analysis compares features with the target variable (Churn) to identify patterns and relationships.

4.2.1 Contract Type vs Churn

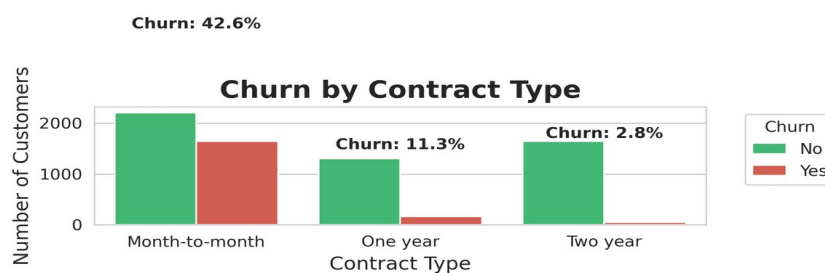


Fig6: **Contract Type vs Churn**

Contract Type	Churned	Total Customers	Churn Rate
Month-to-month	1,643	3,853	42.6%
One year	166	1,473	11.3%
Two year	48	1,695	2.8%

Key Insight: Month-to-month contracts have a 42.6% churn rate, which is 15 times higher than two-year contracts (2.8%). This is the strongest indicator of churn in the entire dataset.

Observation: Customers on flexible month-to-month contracts are much more likely to leave, possibly because they have no financial commitment to stay.

4.2.2 Payment Method vs Churn

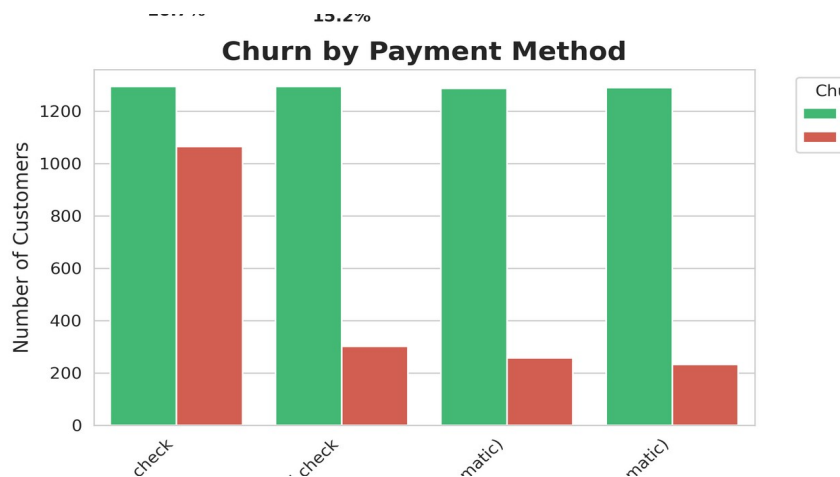


Fig7: Payment Method vs Churn

Payment Method	Churned	Total Customers	Churn Rate
Electronic check	1,065	2,359	45.1%
Mailed check	302	1,596	18.9%
Bank transfer (automatic)	258	1,544	16.7%
Credit card (automatic)	232	1,522	15.2%

Key Insight: Electronic check users have a 45.1% churn rate, nearly 3 times higher than credit card automatic payment users (15.2%).

Observation: Customers using manual or less convenient payment methods are significantly more likely to churn than those with automatic payment methods.

4.2.3 Internet Service vs Churn

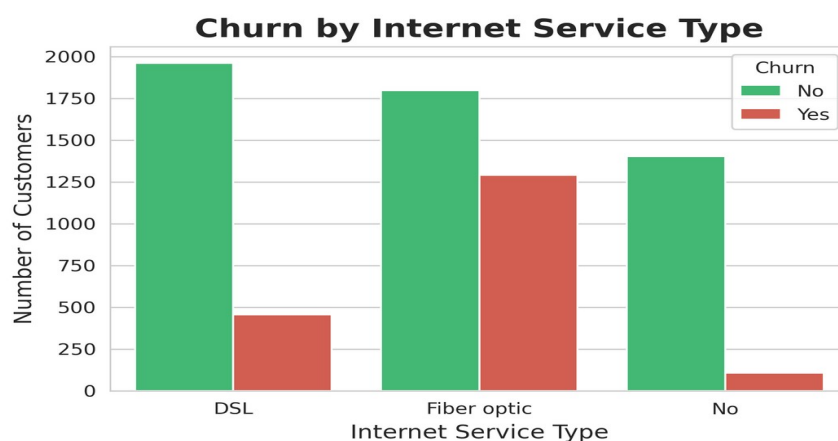


Fig8: Internet Service vs Churn

Internet Service	Churned	Total Customers	Churn Rate
Fiber optic	1,291	3,090	41.8%
DSL	457	2,419	18.9%
No internet	109	1,512	7.2%

Key Insight: Fiber optic users have a 41.8% churn rate, significantly higher than DSL (18.9%) and customers without internet (7.2%).

Observation: Despite potentially being a premium service, fiber optic has the highest churn rate, suggesting possible service quality issues or price sensitivity among fiber optic customers.

4.2.4 Tenure vs Churn (Boxplot)

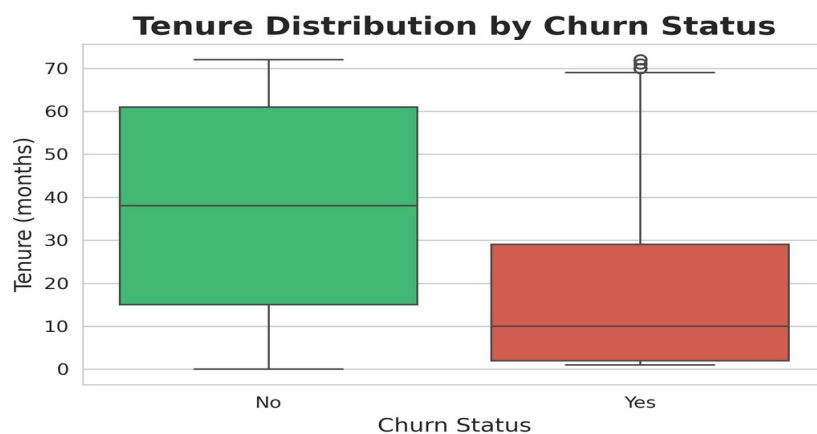


Fig9: Tenure vs Churn Boxplot

Customer Group	Mean Tenure	Median Tenure
Non-Churned	38.44 months	34.00 months
Churned	18.04 months	10.00 months

Key Insight: Churned customers have significantly shorter tenure (about half) compared to non-churned customers. This strongly suggests that customer retention efforts should focus heavily on the first few months of the customer lifecycle.

4.2.5 Monthly Charges vs Churn (Boxplot)

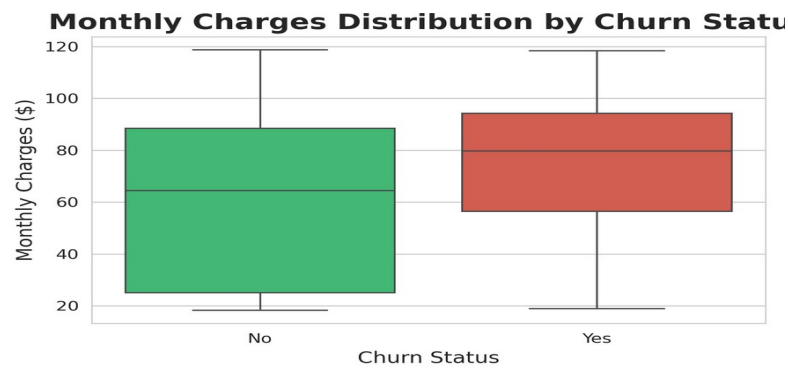


Fig10: Monthly Charges vs Churn Boxplot

Customer Group	Mean Monthly Charges
Non-Churned	\$63.48
Churned	\$68.58

Key Insight: Churned customers have slightly higher average monthly charges (\$5.10 more), suggesting price sensitivity. Customers paying more may have higher expectations and are more likely to leave if those expectations are not met.

4.2.6 Additional Bivariate Findings

Feature	Highest Churn Group	Churn Rate	Lowest Churn Group	Churn Rate
Senior Citizen	Yes	41.6%	No	23.5%
Paperless Billing	Yes	33.4%	No	16.3%
Partner	No	32.8%	Yes	19.7%
Dependents	No	31.2%	Yes	15.5%
Tech Support	No	41.5%	Yes	15.2%

Feature	Highest Churn Group	Churn Rate	Lowest Churn Group	Churn Rate
Online Security	No	41.6%	Yes	14.6%

Observation: Customers who do not have partners or dependents, use paperless billing, or lack add-on services show higher churn rates.

4.3 Correlation Analysis

4.3.1 Correlation Heatmap

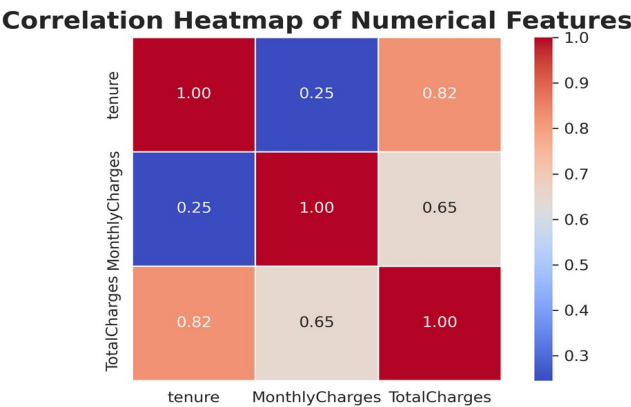


Fig11: Correlation Heatmap

Feature Pair	Correlation Coefficient	Interpretation
tenure & TotalCharges	0.82	Strong positive correlation
MonthlyCharges & TotalCharges	0.65	Moderate positive correlation
tenure & MonthlyCharges	0.25	Weak positive correlation

Key Insights:

1. Customers with longer tenure have significantly higher total charges, which is expected as they have been paying for a longer period.
2. Customers with higher monthly charges accumulate higher total charges over time.
3. There is a weak relationship between tenure and monthly charges, meaning both new and long-term customers can have various price points.

4.4 Scatter Plots

4.4.1 Tenure vs Monthly Charges

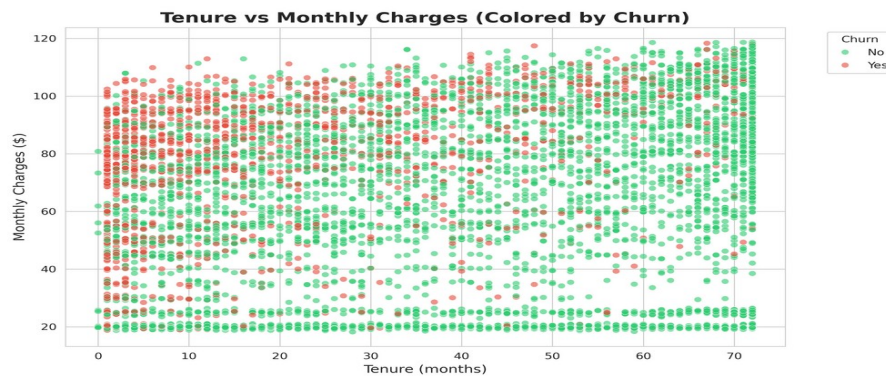


Fig12: Scatter Plot - Tenure vs Monthly Charges

Observation:

1. Customers with shorter tenure show a wider range of monthly charges.
2. Churned customers are more concentrated in the lower tenure range.
3. Correlation between tenure and Monthly Charges: 0.245 (weak positive).

4.4.2 Monthly Charges vs Total Charges

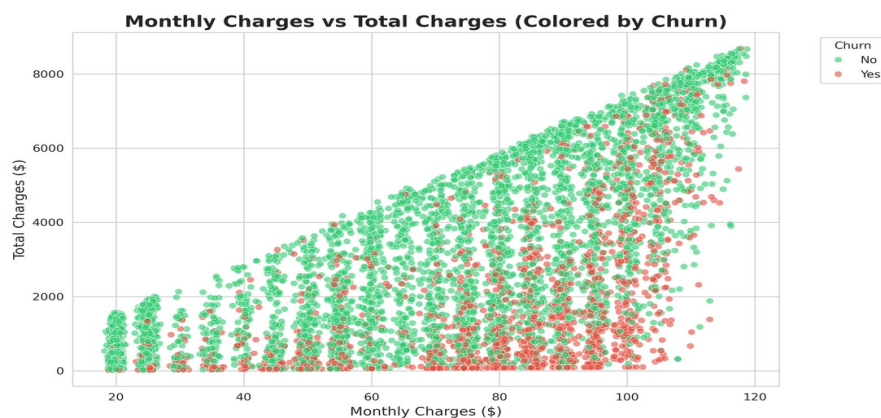


Fig13: Scatter Plot - Monthly Charges vs Total Charges

Observation:

1. Higher monthly charges generally lead to higher total charges.
2. Churned customers are spread across all charge levels, but more concentrated at lower total charges.
3. Correlation between Monthly Charges and Total Charges: 0.650 (moderate positive).

5. Key Findings

Top 3 Factors Affecting Churn

Rank	Factor	Churn Rate	Impact
1	Contract Type (Month-to-month)	42.6%	Highest impact
2	Payment Method (Electronic check)	45.1%	Highest individual risk
3	Internet Service (Fiber optic)	41.8%	Very high risk

Highest Risk Customer Groups

Feature	Highest Risk Group	Churn Rate
Payment Method	Electronic check	45.1%
Contract Type	Month-to-month	42.6%
Internet Service	Fiber optic	41.8%
Senior Citizen	Yes	41.6%
Tech Support	No	41.5%
Online Security	No	41.6%

Lowest Risk Customer Groups

Feature	Lowest Risk Group	Churn Rate
Contract Type	Two year	2.8%
Internet Service	No internet	7.2%
Payment Method	Credit card (automatic)	15.2%
Dependents	Yes	15.5%

Feature	Lowest Risk Group	Churn Rate
Tech Support	Yes	15.2%

Additional Key Insights

1. New customers are at high risk: Churned customers have significantly shorter tenure (18 months vs 38 months for non-churned customers).
2. Price sensitivity matters: Churned customers have slightly higher monthly charges (\$68.58 vs \$63.48).
3. Paperless billing increases risk: Customers with paperless billing have 33.4% churn rate vs 16.3% without.
4. Customer demographics matter: Customers without partners (32.8%) or dependents (31.2%) have higher churn rates.
5. Add-on services reduce churn: Customers with tech support (15.2%) or online security (14.6%) have significantly lower churn rates.

6. Answers to Key Assignment Questions

No.	Question	Answer
1	What percentage of customers churned?	26.45% (1,857 out of 7,021 customers churned). The remaining 73.55% (5,164 customers) are still active.
2	Which customer group has the highest churn?	Electronic check users have the highest churn at 45.1%, followed closely by Month-to-month contract customers at 42.6%, Fiber optic users at 41.8%, Senior citizens at 41.6%, and Customers without tech support at 41.5%.
3	Which numerical features seem related to churn?	Tenure is the most strongly related numerical feature. Churned customers have significantly shorter tenure (mean: 18 months, median: 10 months) compared to non-churned customers (mean: 38 months, median: 34 months). Monthly charges also show a relationship, with churned customers having higher average charges (\$68.58 vs \$63.48).
4	Which categorical features seem related to churn?	Contract Type (Month-to-month: 42.6% vs Two-year: 2.8%), Payment Method (Electronic check: 45.1% vs Credit card auto:

No.	Question	Answer
		15.2%), Internet Service (Fiber optic: 41.8% vs No internet: 7.2%), Senior Citizen (Yes: 41.6% vs No: 23.5%), Paperless Billing (Yes: 33.4% vs No: 16.3%), Tech Support (No: 41.5% vs Yes: 15.2%).
5	Are there missing values or data quality issues?	No. The dataset has 0% missing values in all 21 columns. After removing 22 duplicate rows, the data quality is excellent. Data type issues were resolved during cleaning: TotalCharges was converted to float64 and SeniorCitizen was converted to categorical. All string columns were cleaned of extra spaces.
6	Are there outliers in the dataset?	No. Using the IQR (Interquartile Range) method with a 1.5× multiplier, no outliers were detected in any of the numerical columns (tenure, MonthlyCharges, TotalCharges). The data is clean and suitable for modeling without needing outlier treatment.
7	What are the top 3 factors affecting churn?	1. Contract Type: Month-to-month contracts have 42.6% churn rate (vs 2.8% for two-year contracts) - this is the strongest predictor of churn. 2. Payment Method: Electronic check users have 45.1% churn rate (vs 15.2% for credit card automatic) - this is the highest individual risk factor. 3. Internet Service: Fiber optic users have 41.8% churn rate (vs 7.2% for no internet) - indicating potential service quality or pricing issues.
8	What recommendations can reduce churn?	1. Offer incentives for long-term contracts - Provide discounts or perks for customers switching from month-to-month to yearly contracts. 2. Promote automatic payment methods - Encourage electronic check users to switch to credit card automatic payments through incentives or education. 3. Improve fiber optic service quality - Investigate and address issues causing high churn among fiber optic users. 4. Target senior citizens with retention programs - Create

No.	Question	Answer
		special offers, discounts, or dedicated support for senior citizens. 5. Focus on new customer onboarding - Implement welcome calls, check-ins, and support during the first 6 months. 6. Encourage add-on services - Promote tech support, online security, and other add-ons as they are associated with lower churn rates.

7. Business Recommendations

Based on the comprehensive analysis, the following recommendations are proposed to reduce customer churn:

Short-Term Recommendations (0-3 Months)

Recommendation	Action Items	Expected Impact	Priority
Incentivize long-term contracts	Offer 20% discount for yearly commitment, provide free service upgrades for contract extension, create month-to-month to yearly migration campaign	High	High
Promote automatic payment methods	Offer small discount for auto-pay enrollment, send targeted emails to electronic check users, highlight convenience and reliability of auto-pay	High	High
Target new customers	Implement welcome calls in first week, schedule check-ins at 1, 3, and 6 months, provide onboarding guides and tips	High	High

Medium-Term Recommendations (3-6 Months)

Recommendation	Action Items	Expected Impact	Priority
Improve fiber optic service quality	Conduct quality audits on fiber optic service, increase support staffing for fiber optic issues,	Medium	Medium

Recommendation	Action Items	Expected Impact	Priority
	investigate and address common complaints		
Senior citizen retention programs	Create senior-specific pricing plans, offer personalized support for seniors, provide easy-to-understand billing and service information	Medium	Medium
Service bundle promotions	Bundle add-on services at discounted rates, promote security and support packages to at-risk customers, create loyalty points for service subscriptions	Medium	Medium

Long-Term Recommendations (6+ Months)

Recommendation	Action Items	Expected Impact	Priority
Predictive analytics	Build churn prediction model using machine learning, identify at-risk customers proactively, deploy targeted retention campaigns	Long-term	Medium
Loyalty program	Create tiered loyalty rewards, offer exclusive benefits for long-term customers, recognize customer milestones (1yr, 2yr, etc.)	Long-term	Medium
Customer feedback system	Implement regular Net Promoter Score (NPS) surveys, create feedback loops for service improvement, address pain points identified in surveys	Long-term	Low

8. Conclusion

This exploratory data analysis of the Telco Customer Churn dataset has provided comprehensive insights into the factors driving customer churn. The analysis successfully answers all eight key assignment questions and reveals critical patterns that can inform business strategy.

Key Takeaways

1. Churn is a significant problem with 26.45% of customers leaving the company. This rate is concerning and requires immediate attention.
2. Contract type is the strongest predictor of churn. Customers on month-to-month contracts are 15 times more likely to churn than those on two-year contracts. This single factor presents the biggest opportunity for retention improvement.
3. Payment method and service quality matter significantly. Electronic check users and fiber optic customers show alarmingly high churn rates of 45.1% and 41.8% respectively.
4. New customers are the most vulnerable. Churned customers have significantly shorter tenure (18 months vs 38 months), suggesting that the first year of the customer relationship is critical.
5. The dataset is of high quality. With 0% missing values, no outliers, and clean data types, the analysis results are reliable and actionable.

Final Recommendations

The analysis suggests a focused retention strategy should prioritize:

1. Converting month-to-month customers to long-term contracts through incentives.
2. Migrating electronic check users to automatic payment methods.
3. Improving service quality for fiber optic users.
4. Implementing robust onboarding programs for new customers.
5. Creating targeted retention programs for senior citizens.

Impact

By implementing these data-driven recommendations, the company can significantly reduce churn rates, improve customer retention, and ultimately drive higher revenue and profitability. The insights from this analysis provide a clear roadmap for customer retention efforts, targeting the specific customer segments and behaviors most strongly associated with churn.

List of Visualizations Included

Chart #	Chart Name
1	Churn Count Plot
2	Churn Percentage Pie Chart
3	Tenure Distribution Histogram
4	Monthly Charges Distribution Histogram
5	Senior Citizen Distribution Countplot
6	Contract Type vs Churn
7	Payment Method vs Churn
8	Internet Service vs Churn
9	Tenure vs Churn Boxplot
10	Monthly Charges vs Churn Boxplot
11	Correlation Heatmap
12	Scatter Plot - Tenure vs Monthly Charges
13	Scatter Plot - Monthly Charges vs Total Charges

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Tools Used: Python, Pandas, Matplotlib, Seaborn

Analysis Type: Exploratory Data Analysis (EDA)